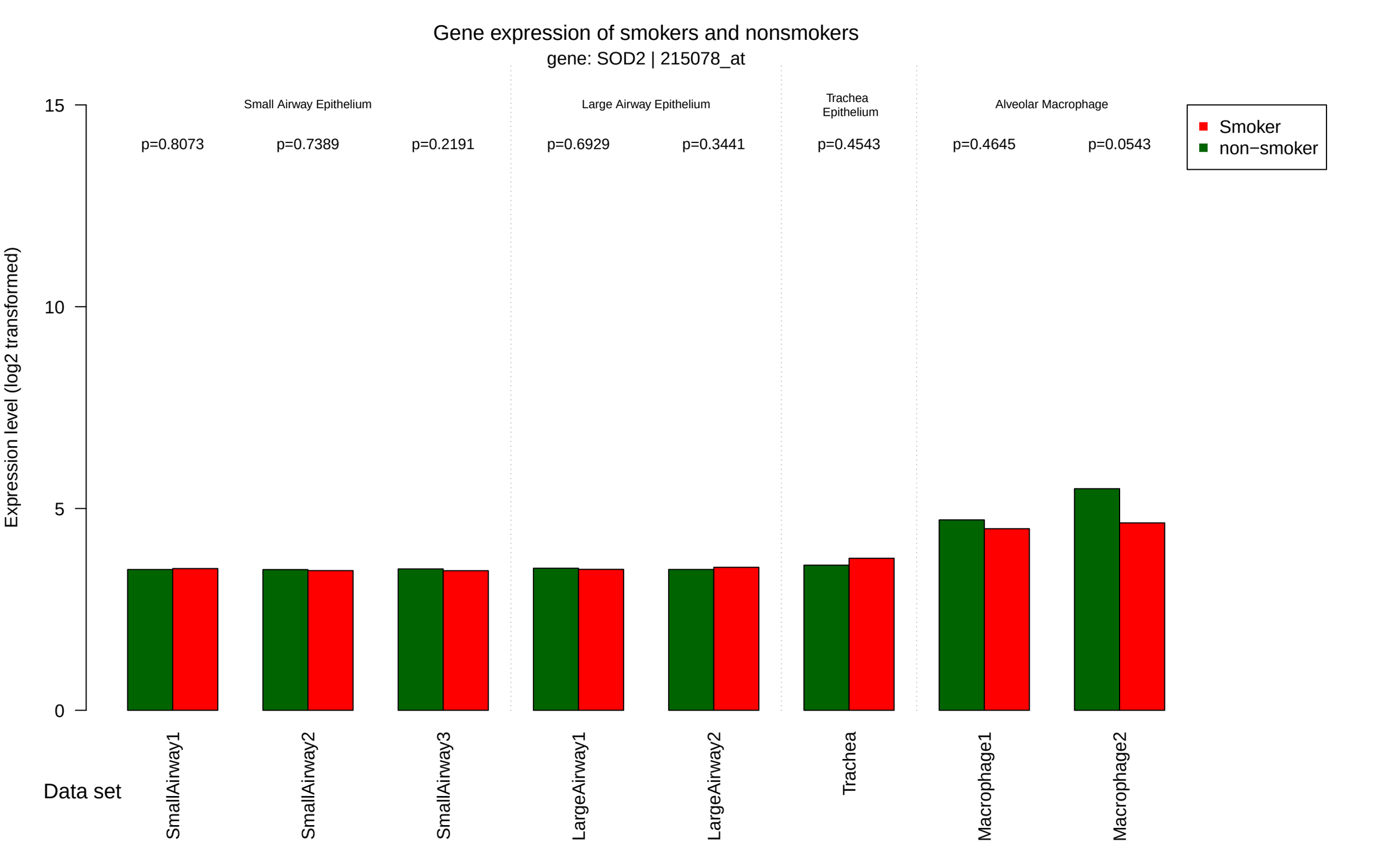
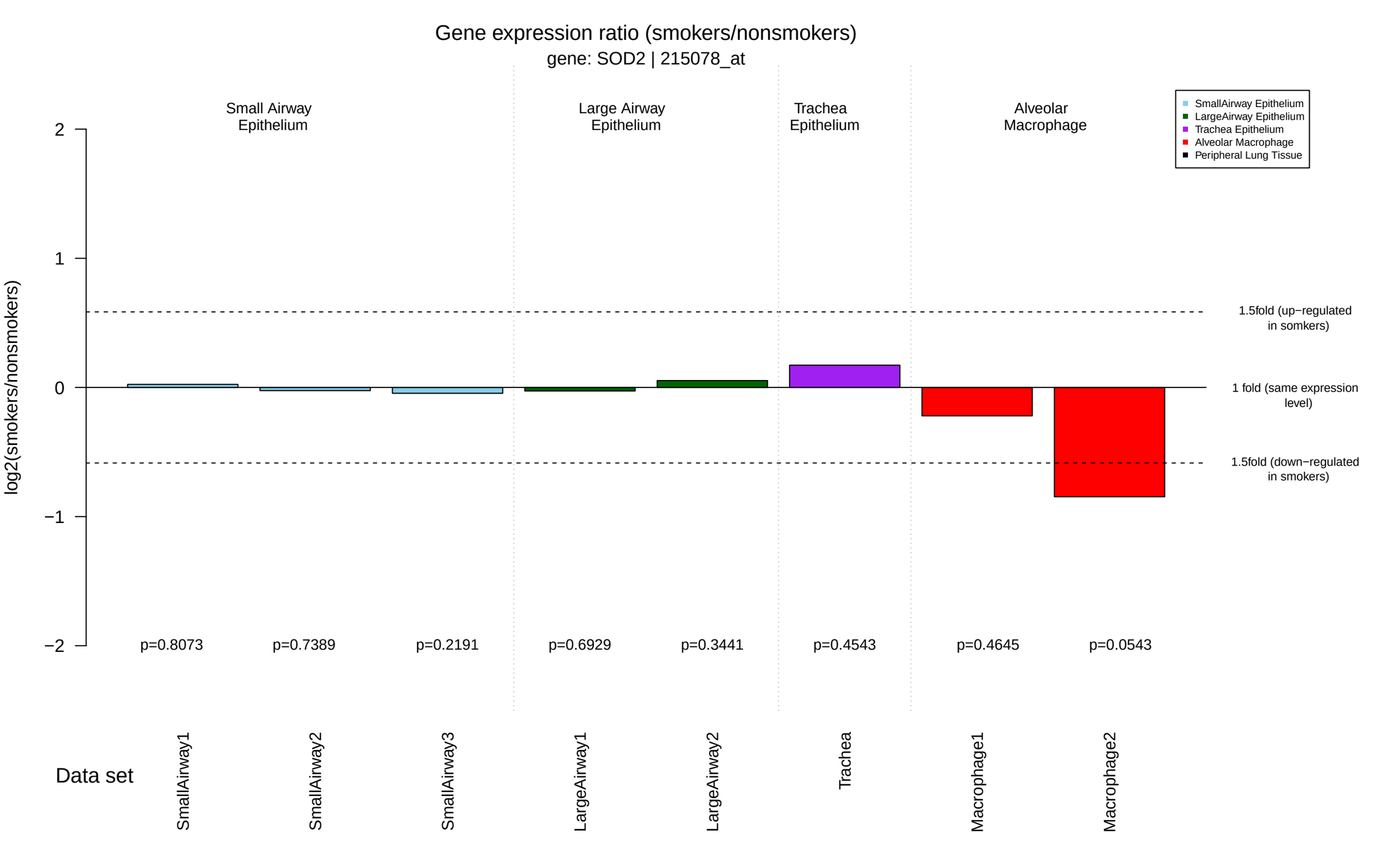
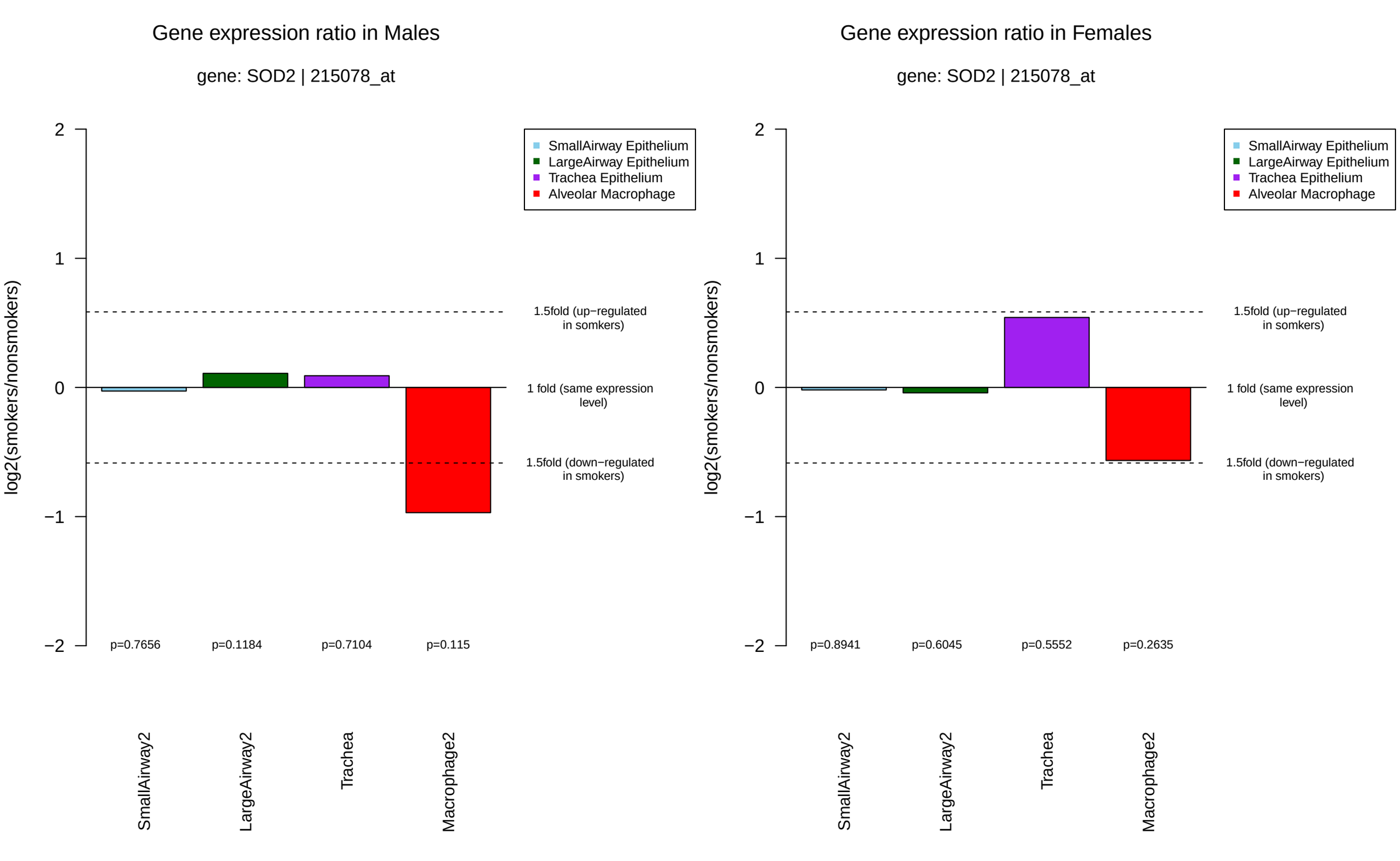
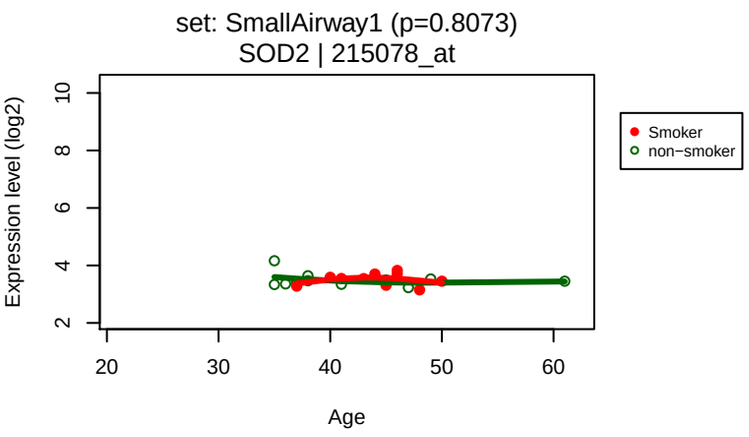
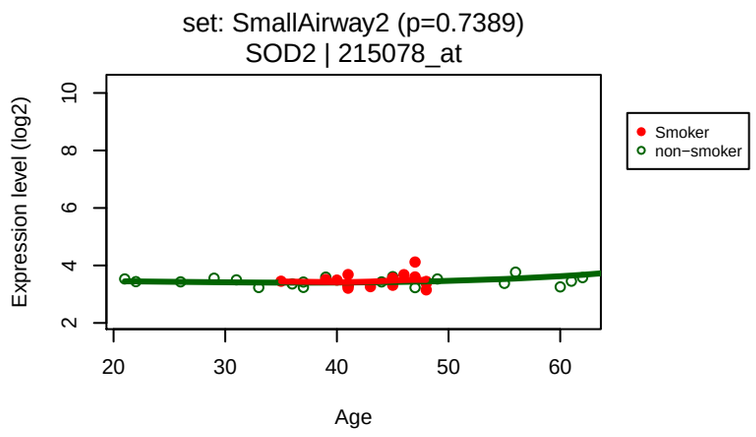
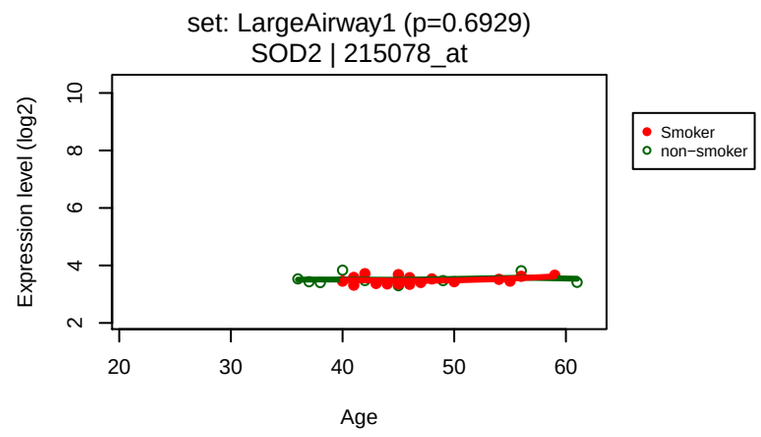

Gene expression of smokers and nonsmokers
gene: SOD2 | 215078_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8073
p=0.7389
p=0.2191
p=0.6929
p=0.3441
p=0.4543
p=0.4645
p=0.0543
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

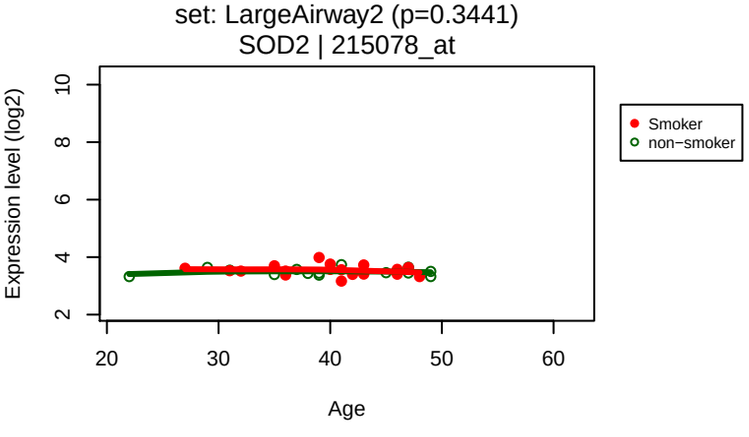

Gene expression ratio (smokers/nonsmokers)
gene: SOD2 | 215078_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8073
p=0.7389
p=0.2191
p=0.6929
p=0.3441
p=0.4543
p=0.4645
p=0.0543
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

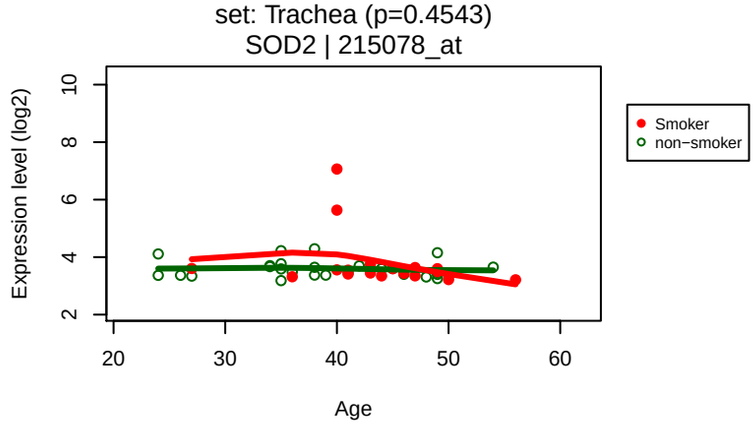

Gene expression ratio in Males
gene: SOD2 | 215078_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7656
p=0.1184
p=0.7104
p=0.115
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOD2 | 215078_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8941
p=0.6045
p=0.5552
p=0.2635
SmallAirway2
LargeAirway2
Trachea
Macrophage2

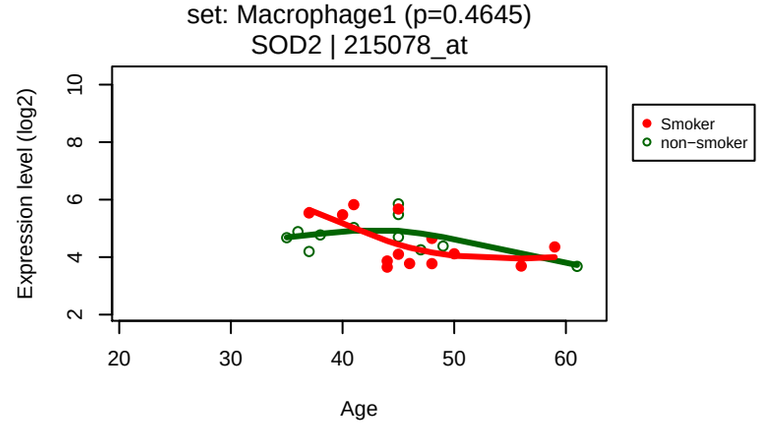

set: SmallAirway1 (p=0.8073)
SOD2 | 215078_at
Expression level (log2)
Age
Smoker
non−smoker

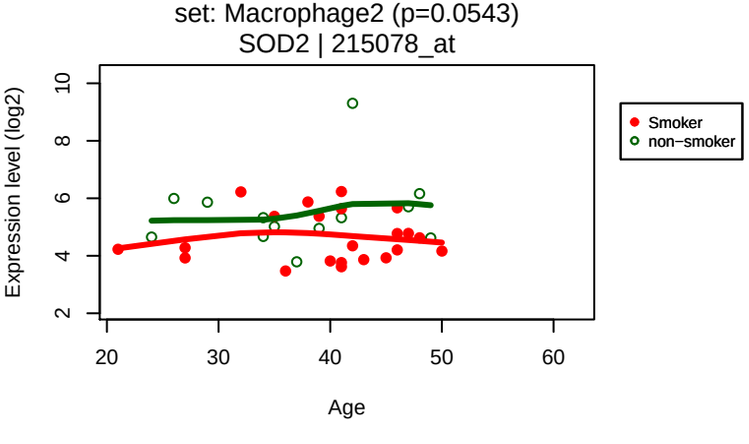

set: SmallAirway2 (p=0.7389)
SOD2 | 215078_at
Expression level (log2)
Age
Smoker
non−smoker

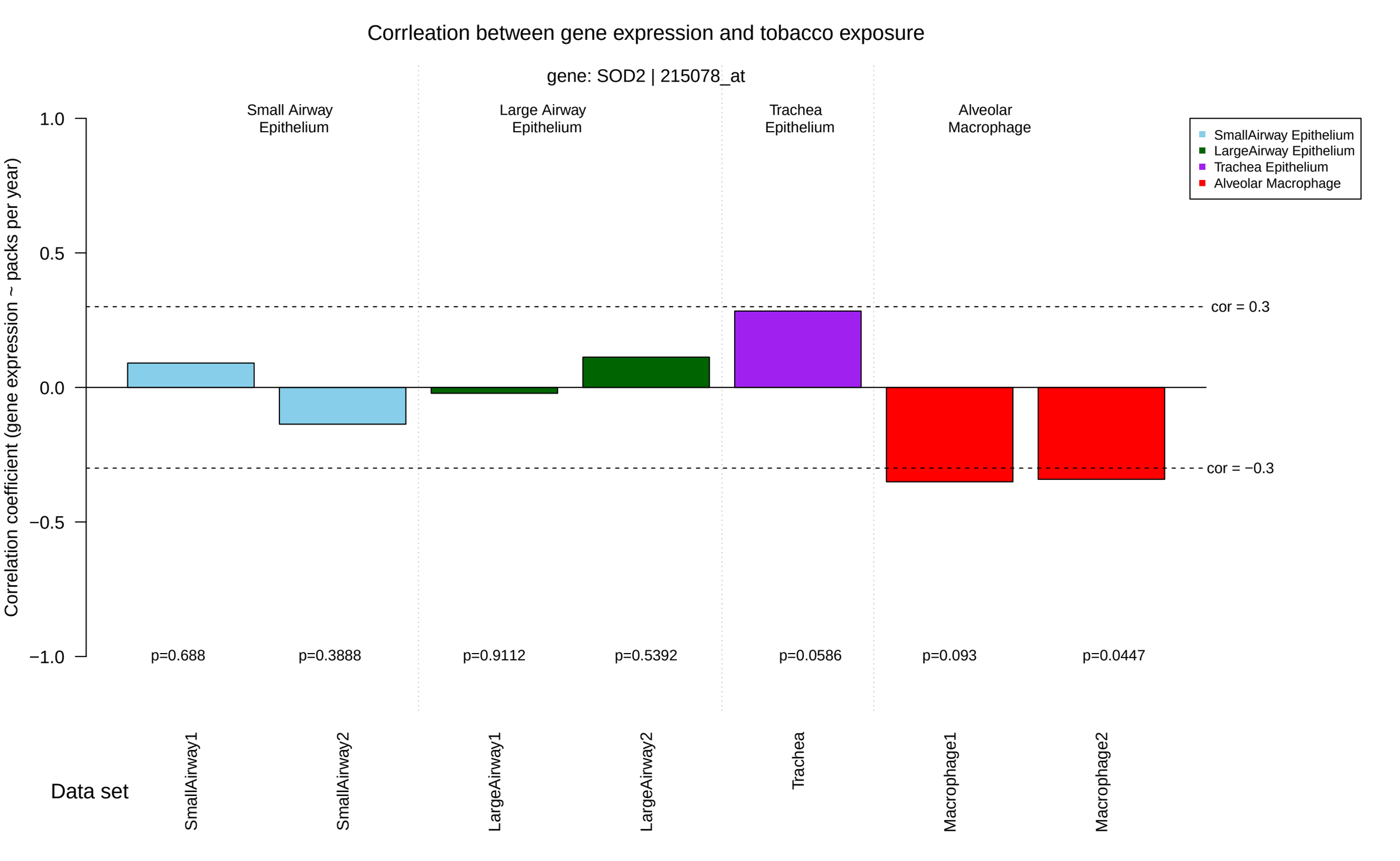

set: LargeAirway1 (p=0.6929)
SOD2 | 215078_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3441)
SOD2 | 215078_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4543)
SOD2 | 215078_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4645)
SOD2 | 215078_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0543)
SOD2 | 215078_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOD2 | 215078_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.688
p=0.3888
p=0.9112
p=0.5392
p=0.0586
p=0.093
p=0.0447
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2